

Document 05: User Flows & Operational Workflows

Project: Natural Intellects School Management System (NISMS)

Version: 1.0

Status: Draft

Purpose:

This document defines the operational workflows and user journeys within the Natural Intellects School Management System.

It explains:

- How users interact with the system.
- The sequence of actions performed.
- Business processes.
- User responsibilities.
- Expected outcomes.

This document serves as the primary reference for:

- Figma design
- Navigation design
- UX design
- Frontend development
- Backend workflows
- System testing

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1. USER LOGIN WORKFLOW

Actors

- Administrator
 - Proprietor
 - Head Teacher
 - Secretary
 - Bursar
 - Teacher
-

Process

User Opens Login Page

↓

Enter Email/Phone

↓

Enter Password

↓

System Validation

↓

Authentication Success

↓

Redirect To Dashboard

Failed Login Process

Invalid Credentials

↓

Display Friendly Error Message

↓

Allow Retry

Requirements

The system shall:

- Support secure authentication.
- Remember active sessions.
- Support password resets.

2. STUDENT REGISTRATION WORKFLOW

Actors

- Administrator
- Secretary

Workflow

Dashboard

↓

Student Management

↓

Add Student

↓

Enter Personal Information

↓

Enter Academic Information

↓

Assign Guardians

↓

Review Information

↓

Save Student

↓

Generate Admission Number

↓

Student Profile Created

Alternative Flow

If admission numbers are manual:

User Enters Admission Number

↓

System Validates Uniqueness

↓

Save Student

Error Conditions

Examples:

- Duplicate admission numbers.
- Missing required fields.
- Invalid dates.

3. STUDENT PROMOTION WORKFLOW

Individual Promotion

Student Profile

↓

Promote Student

↓

Select New Class

↓

Select Stream

↓

Confirm Promotion

↓

Update Active Records

↓

Preserve Historical Records

Bulk Promotion

Select Entire Class

↓

Choose Target Class

↓

Choose Streams

↓

Review Summary

↓

Confirm Promotion

↓

Generate Promotion Report

Business Rules

The system shall:

- Preserve old academic records.
- Preserve old attendance records.
- Preserve financial histories.

4. STUDENT TRANSFER WORKFLOW

Internal Transfer

Example:

P4 Blue

↓

P4 Red

Workflow:

Student Profile

↓

Transfer Student

↓

Select New Stream

↓

Save Changes

External Transfer

Workflow:

Student Profile

↓

Mark As Transferred

↓

Enter Transfer Date

↓

Enter Notes

↓

Archive Student Status

↓

Retain Historical Records

5. TEACHER REGISTRATION WORKFLOW

Actors

- Administrator
 - Secretary
-

Workflow

Teacher Management

↓

Add Teacher

↓

Personal Information

↓

Contact Information

↓

Employment Information

↓

Assign Subjects

↓

Assign Classes

↓

Save Teacher

Business Rules

Teachers:

- Cannot have duplicate staff numbers.

- May teach multiple subjects.
- May teach multiple classes.

6. TEACHER ASSIGNMENT WORKFLOW

Class Teacher Assignment

Workflow:

Select Class

↓

Assign Class Teacher

↓

Save Assignment

Subject Teacher Assignment

Workflow:

Teacher Profile

↓

Assign Subject

↓

Assign Classes

↓

Save Changes

Business Rules

One Class:

↓

One Class Teacher

↓

Multiple Subject Teachers

7. ATTENDANCE WORKFLOW

Actors

- Teachers
 - Administrators
-

Daily Attendance Workflow

Teacher Login

↓

My Classes

↓

Select Class

↓

Mark Attendance

↓

Review Records

↓

Submit Attendance

Attendance Status Options

Students may be:

- Present
 - Absent
 - Sick
 - Excused
-

Administrative Workflow

Dashboard

↓

Attendance Reports

↓

Daily Reports

↓

Weekly Reports

↓

Monthly Reports

Business Rules

Attendance shall:

- Be recorded once per day.
 - Support editing by administrators.
 - Preserve historical records.
-

8. MARKS ENTRY WORKFLOW

Actors

- Teachers
-

Workflow

Teacher Dashboard

↓

Select Class

↓

Select Subject

↓

Select Term

↓

Enter Marks

↓

Review Entries

↓

Submit Marks

Validation Rules

Marks:

- Must be numeric.
 - Cannot exceed maximum values.
 - Cannot be negative.
-

9. REPORT CARD GENERATION WORKFLOW

Actors

- Head Teacher
 - Administrator
-

Individual Reports

Workflow:

Student Profile

↓

Generate Report

↓

Select Term

↓

Generate PDF

↓

Print Report

Bulk Reports

Workflow:

Class Reports

↓

Select Class

↓

Select Term

↓

Generate Reports

↓

Download PDF Package

Business Rules

Reports:

- Cannot change after final approval.
- Preserve historical information.

10. SCHOOL FEE PAYMENT WORKFLOW

Actors

- Bursar

Payment Workflow

Student Profile

↓

Fees Module

↓

Record Payment

↓

Select Payment Method

↓

Enter Reference

↓

Save Transaction

Supported Methods

- Cash
 - Mobile Money
 - Bank Deposits
-

Business Rules

Payments:

- Cannot be deleted.
 - May only be adjusted through correction entries.
-

11. RECEIPT GENERATION WORKFLOW

Workflow

Payment Recorded

↓

Generate Receipt

↓

Assign Receipt Number

↓

Print Receipt

↓

Store Digital Copy

Receipt Contents

Include:

- Student name
- Class

- Amount paid
- Payment method
- Date
- Reference number

12. ANNOUNCEMENT WORKFLOW

School-Wide Announcements

Workflow:

Dashboard

↓

Announcements

↓

Create Announcement

↓

Enter Content

↓

Select Audience

↓

Publish

Class Announcements

Possible Audiences:

- Entire school
 - Specific classes
 - Teachers
-

Future Possibilities

Potential support:

- SMS notifications
- WhatsApp notifications

13. WEBSITE CONTENT MANAGEMENT WORKFLOW

Actors

- Administrators

News Management

Workflow:

Website Dashboard

↓

News Management

↓

Create News Article

↓

Upload Images

↓

Publish

Gallery Management

Workflow:

Gallery

↓

Upload Images

↓

Add Descriptions

↓

Publish

Admissions Content

Workflow:

Admissions Page

↓

Edit Information

↓

Upload Documents

↓

Save Changes

14. ERROR HANDLING PRINCIPLES

User-Friendly Messages

Avoid:

"Database Exception Error"

Prefer:

"Admission Number Already Exists"

Recovery Actions

Users should:

- Correct errors easily.
 - Retry failed operations.
 - Preserve unsaved work where possible.
-

Validation Before Submission

The system shall:

- Validate forms early.
 - Prevent invalid submissions.
 - Provide clear guidance.
-

STRATEGIC CONCLUSIONS

The Natural Intellects School Management System shall prioritize:

- Simple workflows.
- Minimal user confusion.
- Mobile-friendly operations.
- Preservation of historical records.
- Fast access to common tasks.

These workflows shall serve as the foundation for:

- Figma wireframes
- Navigation structures
- Frontend implementation
- Backend business logic
- Cloud AI development prompts

End of Document 05.